

# Milestone 4: Advanced AI Integrations, Coding Arena & System Validation

## 1. Implemented Features: Google Gemini AI Integration

- **Gemini Configuration Service:** Formulated GeminiAiService.java. Handles connection templates mapping to the Gemini API gateway endpoints (gemini-1.5-flash).
- **AI Study Buddy & Assistant:** Configured conversational client modules. Explains engineering concepts, maps learning paths, and answers user questions.
- **Personalized Learning Recommendations:** Implemented algorithms parsing user progress data and matching it with suitable course suggestions.
- **AI Career Roadmaps:** Developed generators outputting step-by-step career path milestones and skill requirements based on career goals.
- **AI Resume Suggestions:** Programmed document verification services reviewing resumes and suggesting optimization adjustments.

## 2. Implemented Features: Intelligent Career Matching & Analytics

- **Career Recommendation Engine:** Designed algorithms parsing student profiles and calculating matching metrics against active job opportunities.
- **Skill Matching Calculations:** Implemented cosine similarity algorithms comparing student skill vectors with job posting requirements.
- **Smart Progress Insights:** Created analytics modules analyzing study patterns to suggest optimal learning schedules.

## 3. Implemented Features: Resume Builder & Coding Arena Sandbox

- **Resume Builder Module:** Programmed form wizard templates gathering career details. Styled printable document formats with PDF download support.
- **Coding Arena UI Layout:** Integrated Monaco Editor widget supporting themes, syntax highlighting, and auto-formatting options.
- **Compilation Sandbox:** Deployed execute endpoint (/api/sandbox/execute) passing code inputs to safe runners and verifying them against testing parameters.
- **Automated Assertion Framework:** Configured assertion controllers validating execution returns and returning test results to the editor view.

## 4. End-to-End System Testing & Deployment

- **JUnit Testing Suite:** Built test cases validating controllers, security filters, and user registration flows.
- **Verification Workflows:** Tested integrated workflows (Login -> Student Dashboard -> Course Catalog -> Course Enrollment -> Resume Creation -> Matching Job Recommendations).
- **Deployment Configuration:** Formulated Dockerfile and environment configs for production deployments.

## 5. Deliverables & Technical Verification

- **[X]** Recommendation controllers returning matched positions under `/api/recommendations/jobs`.
- **[X]** Resume Builder wizard form with document preview layouts.
- **[X]** Coding Arena text editor module and assertion verification engine.
- **[X]** AI study helper chat interface connected to Google Gemini model endpoints.